

Physics :-

Temp symbol is K (not T)
Base quantities are fundamental

Q:2 Attempt the following questions.

Q.(i)

What are base quantities? Write any 4 base quantities with their SI units and symbol?

A(i)

Base Quantities:-

Base or fundamental physical quantities are such as (time, length, temperature), these are quantities can be derived from other physical quantities.

SI units and symbols:-

name	SI unit	symbols
Time	seconds	s
Length	metre	m
Temperature	Kelvin	K
Mass	Kilogram	kg

Q2(ii): 2/3

Q. (i) (or) $0.00072 \text{ m} = 7.2 \times 10^{-4} \text{ m}$ (OK)
Prefix should be 0.72 mm

Ans (i) (or) OR prefixes OK; $3.6 \times 10^3 \text{ g} = 3.6 \text{ kg}$

Derived Quantities:-

Derived quantities are those physical quantities which can be obtained by multiplying or dividing base quantities.

Examples:-

- * Area $\rightarrow$ meter cubic meter
- * Speed velocity $\rightarrow$ meter per second
- * Acceleration $\rightarrow$ meter per second second cubic.

Q. (ii)

Ans (ii)

Scientific notation:-

0.00072 m

$7.2 \times 10^{-4} \text{ m}$

7.2 mm

Q. (ii) (or)

Ans (ii) (or)

$10^{-3} = \text{milli} \rightarrow \text{m}$

$10^6 = \text{Mega} \rightarrow \text{M}$

Conversion:-

Scalars CAN be added.
Area is scalar (m^2).

$$3.6 \times 10^3 \times 10^3$$

$$10^3$$

$$3.6 \times 10^3 \times 10^3$$

$$3.6 \times 10^6$$

$$3.6 \times 10^6 \text{ kg}$$

Q(iii)

Ans (iii)

Scalars

Vectors

* Scalar quantities are those quantities which can only be have numerical magnitude (with a proper unit) not a direction.

vector quantities are those quantities which can have both numerical magnitude with a proper unit and direction

* Scalar quantities can be not be added, subtracted or multiplied by an ordinary algebra

vector quantities can be added, subtracted or multiplied by a vector algebra.

* Example:- Length,
time,
mass

Example:- Force
Area
Acceleration.

Need clear steps: place tail of 2nd at head of 1st; resultant = tail(first) to head(last).

Q. 2.iii (or)

Ans. iii (or)

Adding vectors using head to tail rule:-

The process of combining two vectors and getting a result called resultant. The addition of vectors is not about summing vectors and get the simple. It is about the precise measurement using scale. The process of using head to tail rule is that the first head of the vector should be in the tail of the last vector, causing resultant.

Q. iv)

Ans (iv)

Least Count :-

Least count can be obtained by dividing the total number of pitch with the total numbers of division.

$$VSD = 9 \times 1 \text{ mm}$$

$$10$$

$$= 0.9 \text{ mm} - 1 \text{ mm}$$

$$0.1 \text{ mm}$$

Q: iv (or)

Ans (iv) (or)

Pitch :- 1 mm

Divisions :- 100

L.C = ?

$$L.C = \frac{\text{number of pitch}}{\text{total number of divisions}}$$

$$= \frac{1}{100}$$

$$= 0.01 \text{ mm}$$

Q2(v): 1.5/3

Define zero error: non-zero reading when anvil/spindle just touch.
Correction: subtract/add zero error.

Q (v)

Ans (v)

Turn the ~~thimble~~ ~~thimble~~ by thimble until anvil and spindle meet. The datum line of the ~~screw~~ ^{screw} gauge must be equal to the circular main scale. If the main scale and datum line doesn't meet ~~opposite~~ ^{opposite} there may be a zero error.

Random error not always negative.
Give 1 clear reduction method for each.

Q: v (or)

Ans: v (or)

Systematic error

* Systematic error can be happen regularly

* They may be negative or positive.

* Types of systematic errors are:-

- 1) Personal error
- 2) Imperfection errors
- 3) External error.

Random error

Random error can be unpredictable, happened irregularly.

They are negative.

Q2(vi): 1/3

Rules incomplete.
Rounding: 7.8465 $\sim$ 7.85 (2 d.f.)
Two decimals: 7.85

by experim
experimentation,
imperfection of
methods

Q: vi

Ans: vi

Three rules of significant figures:-

* All non zero digit are directly reported to significant figures.

* All non zero digits (from 9 to 1) are reported as significant figure

* If 0 comes first after decimal then it is directly reported to count digits after decimal digit for significant figures

————— Q. (vi) (or) —————

————— Answer (vi) (or) —————

♥ Three significant figures :-
* 7.8465

7.84

♥ Decimal places :-

7.84

————— Q (vii) —————

————— Answer (vii) —————

Q2(vii): 3/3

* Distance :-

The path ^{covered} between the initial path to final path is called distance.

* Displacement :-

The shortest ^{path} covered between the initial point to final point is called displacement.

Displacement can be zero :-

Displacement can be zero while distance is not because when an object moves from one path to another its distance occurs but when if an object starts and ending at the same point then displacement is zero.

Q2(vii) OR: 2.5/3

difference: avg speed uses distance
avg velocity uses displacement
direction: zero

Q: vii (or)

Ans: (vii) (or)

Average speed :-

Average The total distance covered with the passage of time is called average speed. Its mathematical form is or equation is :-

Average speed = ?

$$v = \frac{s}{t}$$

Average velocity :-

The total displacement covered with the passage of time is called average velocity (denoted by $\vec{v}$). Its mathematical equation is :-

$$\vec{v} = \frac{\vec{s}}{t}$$

Differences :-

Average Speed	Average Velocity
* It's is denoted by v .	It is denoted by $\vec{v}$.
* It doesn't use direction	It includes direction too.
* It's a scalar quantity	It's a vector quantity.

Q2(viii): 1.5/3

Uniform acceleration = constant acceleration (not uniform velocity)

Q: viii

Answer (vii)

Acceleration :-

Acceleration is a measured of velocity $\Delta \vec{v}$ with the passage of time Δt .
Mathematically :-

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = \vec{a} = \frac{v_f - v_i}{t_f - t_i}$$

Uniform Acceleration	Non-uniform acceleration
* Uniform acceleration are when the body is moving in uniform velocity.	Uniform non-uniform acceleration are when the body is moving

$$g = 9.8 \text{ m/s}^2 \text{ (unit)}$$

viii (or)

viii (or)

Motion due to gravity:-

When freely falling objects falling with acceleration then it is called motion due to acceleration ($\vec{a}$) gravity. The value of 'g' near earth's surface with unit.

$$g = 9.81 \text{ m/s}^2$$

(x) (or)

Q2(x): 3/3

(x) (or)

Two conditions of Equilibrium:-

1st condition of Equilibrium:-

When vectors sum of all forces acting on a body is zero then the first condition is satisfied. Mathematically if $F_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 \dots = \vec{F}_{\text{net}}$

Mathematically:-

$$F_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 \dots = F_{\text{net}}$$

$$F_{\text{net}} = 0$$

$$\Sigma = 0$$

2nd condition of Equilibrium:-

When vectors of sum of all torques acting on a body is zero then the second

condition of equilibrium is satisfied
Mathematically, it is if $\tau_{net} = \tau_1 + \tau_2 + \tau_3 = 0$
- τ_{net}

Q2(XI): 2.5/3

$$\tau_{net} = 0$$
$$\tau_{net} = 0$$

(xi) (first option)

Law describes 'what', theory explains 'why'.

Answer

Hypothesis:-

A hypothesis is a tentative idea made before any research or assumption. It can be verified by further investigation.

Theory:-

The theory is a description of nature that scientists have verified through rigorous testing. It can be true or false after experimentation and implementation.

Law:-

A law is a statement that observes regularity and lawful statement.

Theory	Law
A Theory is description of nature that scientists have verified through rigorous testing.	A law is statement that is made and observes regularity.

Need principle of moments statement
+ example.
About pivot $\Sigma(\text{anticlockwise moments}) = \Sigma(\text{clockwise moments})$

LONG QUESTION

Q.

Q:5

A:5

Equilibrium:-

Equilibrium is stated that all individual forces, moment of forces and torques of forces upon an object are unbalanced. All net forces and moment of forces are zero an object is said to be in Equilibrium.

Conditions of Equilibrium:-

1st Conditions :-

Vector of sum of all forces acting on a body together then the first condition of Equilibrium is satisfied, if F_{net} is sum of all forces $F_1 + F_2 + F_3$. Then mathematically,

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots = \vec{F}_{net}$$

$$F_{net} = 0$$

$$\Sigma = 0$$

2nd Condition:-

Vector of sum of all torques acting on a body together then the second condition of Equilibrium

is satisfied. If $\tau_{\text{net}} = \tau_1 + \tau_2 + \tau_3 \dots$ is sum of all torques $\tau_1 + \tau_2 + \tau_3 \dots$ then mathematically,

$$\tau_{\text{net}} = \tau_1 + \tau_2 + \tau_3 \dots = \tau_{\text{net}}$$

$$\tau_{\text{net}} = 0$$

$$\Sigma = 0$$

Types of Equilibrium:-

* Static equilibrium:-

When an object is at rest under the actions of several forces or under the actions of several torques then the body is said to be in static equilibrium.

Add a numerical/example (sodaaw/balence) for full marks.

For Example:-

A book is resting on the table.

* Dynamic Equilibrium:-

When a body is moving at uniform velocity under the action of forces and torques then the body is said to be in dynamic equilibrium. It is further divided in two sub types.

* Sub types of dynamic equilibrium:-

Translational dynamic equilibrium:-

When a body moves at uniform velocity, then the object is said to be in translational dynamic equilibrium. For example, a paratrooper is falling down in linear velocity.

Rotational dynamic equilibrium:-

When a body moves at uniform rotational velocity, then the object is said to be in rotational dynamic equilibrium. For example; a fan is on ceiling rotating in a rotating uniform velocity.

Principle moment :-

The second condition of equilibrium is called principle of moment.

Principle of moment:- In equilibrium for an object in equilibrium, the sum of the clockwise moment taken about the pivot must be equal to the sum of anticlockwise moment taken about the same pivot.